

MODEL EXAM

PART-A

1. State Cayley-Hamilton theorem.
2. Two of the eigen values of $A = \begin{pmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{pmatrix}$ are 3 and 6. Find the eigen values of A^{-1} .
3. Prove that $\nabla(r^n) = nr^{n-2}\vec{r}$.
4. Calculate curl $\vec{F}$, if $\vec{F} = xyz\vec{i} + 3x^2y\vec{j} + (xz^2 - y^2z)\vec{k}$ at (1,-1,1).
5. Define Laplace transform.
6. State initial and final value theorems on Laplace transform.
7. State Convolution theorem on Laplace theorem.
8. Find $L^{-1} \left[\frac{s^2 - 3s + 4}{s^3} \right]$.
9. State the Fourier integral theorem.
10. State any two properties of Fourier transform.
11. Find eigen values of A^{-1} , where $\begin{pmatrix} 1 & 2 \\ 0 & 4 \end{pmatrix}$.
12. Write the matrix of quadratic form $2x^2 + 8z^2 + 4xy + 10xz - 2yz$.
13. Show that the vector $\vec{F} = 3y^4\vec{i} + 4x^3z^2\vec{j} - 3x^2y^2\vec{k}$ is solenoidal.
14. Show that $\vec{F} = (6xy + z^3)\vec{i} + (3x^2 - z)\vec{j} + (3xz^2 - y)\vec{k}$ is irrotational.
15. Find $L[5^t]$.
16. Define periodic function $f(t)$.
17. State Convolution theorem on Laplace theorem.
18. Find $L^{-1} \left[\frac{1}{(s+1)^2} \right]$.
19. State the Convolution theorem for Fourier transform.
20. Find the Fourier sine transform of $\frac{1}{x}$.

PART-B UNIT-I

1. Find the eigen values and eigen vectors of the matrix $A = \begin{pmatrix} 2 & -2 & 3 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{pmatrix}$.
2. Verify Cayley-hamilton theorem for the matrix $A = \begin{pmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$ and also find the inverse. Also find A^4 .
3. Reduce the quadratic form $6x_1^2 + 3x_2^2 + 3x_3^2 - 4x_1x_2 - 2x_2x_3 + 4x_3x_1$ into canonical form by orthogonal transformation.
4. Find the eigen values and eigen vectors of the matrix $A = \begin{pmatrix} 3 & -4 & 4 \\ 1 & -2 & 4 \\ 1 & -1 & 3 \end{pmatrix}$.
5. Verify Cayley-hamilton theorem for the matrix $A = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 2 \end{pmatrix}$ and also find the inverse. Also find A^4 .
6. Reduce the quadratic form $3x^2 + 5y^2 + 3z^2 - 2xy - 2yz + 2zx$ into canonical form by orthogonal transformation.

UNIT-II

1. Verify Gauss divergence theorem for $\vec{F} = (2x - z)\vec{i} + x^2y\vec{j} - xz^2\vec{k}$ taken over the cube bounded by $x=0$, $x=1$, $y=0$, $y=1$, $z=0$ and $z=1$.
2. Verify Stokes theorem for $\vec{F} = y^2z\vec{i} + z^2x\vec{j} + x^2y\vec{k}$, where S is the open surface of the cube formed by the planes $x = \pm 1$, $y = \pm 1$, $z = \pm 1$ in which plane $z = -1$ is open.
3. Verify Gauss divergence theorem for $\vec{F} = (x^2 - yz)\vec{i} + (y^2 - zx)\vec{j} + (z^2 - xy)\vec{k}$ taken over the rectangular parallelepiped bounded $0 \leq x \leq a$, $0 \leq y \leq b$ and $0 \leq z \leq c$.
4. Verify Stokes theorem for $\vec{F} = (x^2 + y^2)\vec{i} - 2xy\vec{j}$, integrated around the rectangle bounded by the lines $x = \pm a$, $y = 0$ and $y = b$.

UNIT-III

1. Verify Initial value theorem and final value theorems for the function $f(t) = e^{-t}(t + 2)^2$.
2. Find the Laplace transform of $f(t) = \begin{cases} t, & \text{for } 0 \leq t \leq a; \\ 2a - t, & \text{for } a \leq t \leq 2a. \end{cases}$ and $f(t + 2a) = f(t)$ for all t .

3. Find the Laplace transform of $f(t) = \begin{cases} -E, & \text{for } 0 < t < \pi; \\ E, & \text{for } \pi < t < 2\pi. \end{cases}$
4. Evaluate (i). $L[e^{at}t^n]$, (ii). $L[e^{at}\cos bt]$
5. Verify Initial value theorem and final value theorems for the function $f(t) = t^2e^{-4t}$.
6. Find the Laplace transform of $f(t) = \begin{cases} t, & \text{for } 0 \leq t \leq a; \\ 2a - t, & \text{for } a \leq t \leq 2a. \end{cases}$ and $f(t + 2a) = f(t)$ for all t .
7. Find the Laplace transform of $f(t) = \begin{cases} K, & \text{for } 0 < t < a; \\ -K, & \text{for } a < t < 2a. \end{cases}$
8. Evaluate (i). $L[te^{-2t}\cos 3t]$, (ii). $L[t^2\sin at]$

UNIT-IV

1. Find $L^{-1}\left[\frac{5s^2 - 15s + 7}{(s+1)(s-2)^3}\right]$
2. Using Convolution theorem $L^{-1}\left[\frac{s^2}{(s^2+a^2)(s^2+b^2)}\right]$.
3. Solve using Laplace theorem, $\frac{d^2x}{dt^2} + 3\frac{dx}{dt} + 2x = e^{-t}\sin t$, given that $x(0) = 0$
and $x'(0) = 1$.
4. Solve $\frac{d^2y}{dt^2} + \frac{dy}{dt} = t^2 + 2t$, where $y(0) = 4$ and $y'(0) = -2$.
5. Find $L^{-1}\left[\frac{1}{(s+1)(s^2+4)}\right]$.
6. Using Convolution theorem $L^{-1}\left[\frac{s^2}{(s^2+4)(s^2+9)}\right]$.
7. Solve using Laplace theorem, $\frac{d^2y}{dx^2} + 9y = \cos 2x$; given that $y(0) = 1$ and $y\left(\frac{\pi}{2}\right) = -1$.
8. Solve $y'' - 3y' + 2y = e^{2t}$, given that $y(0) = 3$; $y'(0) = 5$. Using Laplace transform.

UNIT-V

1. Find Fourier transform of $f(x) = \begin{cases} a^2 - x^2, & |x| < a \\ 0, & \text{if } |x| > a. \end{cases}$ and hence deduce that

$$(i). \int_0^{\infty} \frac{\sin x - x \cos x}{x^3} dx = \frac{\pi}{4}. (ii). \int_0^{\infty} \left[\frac{\sin x - x \cos x}{x^3} \right]^2 dx = \frac{\pi}{15}.$$

2. Find the Fourier cosine and sine transform of e^{-ax} ; $a > 0$ and deduce inversion formula and parsevals identity formula.

3. Find Fourier transform of $f(x) = \begin{cases} 1 - |x|, & \text{for } |x| \leq 1; \\ 0, & \text{for } |x| > 1. \end{cases}$ and hence deduce that

$$\int_0^{\infty} \left(\frac{\sin t}{t} \right)^4 dt = \frac{\pi}{3}.$$

4. Find the Fourier cosine and sine transform of e^{-ax} ; $a > 0$ and deduce inversion formula.

5. Prove that $e^{-\frac{x^2}{2}}$ is self reciprocal under Fourier transform.